Process Control

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Introduction

- Used in industries and manufacturing
- Process control is used in automation processes such as material handling, state machines and sequencing .
- It controls variables such as temperature, pressure and flow
- Improves quality of product and requires fewer personal

Theory

- In order to keep the water levels constant, the flowrate by the pump must be identical to the flowrate that is leaving vessel via CV2 and the solenoid valve.
- The level of the water was to remain constant whilst controlling the speed of the pump because the flow of water entering the vessel is proportional to the pump speed controlled manually on the mimic diagram of the computer
- Rather controlling the process manually, the computer is automated to control the water levels to attain equilibrium through both of the valves.
- Given at different parameters, the computer will control how much water is being pushed in and out.

Lab Components

- Solenoid Valve
- CV1
- CV2
- CV3
- Sump Tank
- PCT50
- Ball Valve

Material and Supplies

- DI Water (blue hose on the sink)
- PC with PCT50 software loaded
- PCT50 Level Process
- USB Cable
- Must be near outlet power plug for the electrical equipment

Procedure

This lab was broken down into 2 parts.

- Part A - Manually controlled the speed of the pump
- Part C - Automation using P only Controller and P+I Controller

Both experiments start by filling the vessel with RO water. Optimal level is 30 mm below the cut out.

Procedure for experiment a

1. Fully open outlet valve CV2 and fully open inlet valve CV1
2. **Important- if the level on the mimic diagram is not 0mm; click the zero button.**
3. Gradually increase the speed of the pump unitil the input equals the output. This can be observed when the levels remain constant.
4. Once it is constant open solenoid valve and allow the levels to fall 20mm
5. Once the level reached 20mm from starting point, ramp up the speed to achieve levels higher than 20mm from starting point
6. Close solenoid valve again and allow the water level to decrease by 20mm again.
7. Using a new sheet for data logging, begin test by gradually increasing the pump speed until water reaches 150mm.
8. Once level reached 150mm turn off pump and allow the vessel to empty
9. Once the vessel is empty quickly turn pump to 75% speed and allow the vessel to fill again to the 150mm mark.
10. Allow the vessel to empty again

Procedure for experiment c

Procedure for P- only controller

1. Select PID box; set Proportional Band P to 200%; Integral Time I to 0; Derivation time D to 0; Set Point to 75m; Click apply!
2. The pump will start and will gradually rise and water will drain from the vessel.
3. Open the solenoid valve and observe the level fall slightly .
4. Adjust the setpoint to 100 mm and observe the response. Return to 75mm
5. Continue to Adjust the P as following and repeat steps 1-4
   a. 50%
   b. 20%
   c. 10%
   d. 5%
   e. 2%
6. End of test

Procedure for experiment c

Procedure for -P+I only controller

1. Select PID box; set Proportional Band P to 200%; Integral Time I to 0; Derivation time D to 0; Set Point to 75m; Click apply!
2. Adjust the I term to 100 seconds in the PID controller.
3. When the level is settled, open the solenoid valve to disturb the process.
4. Adjust setpoint to 100 mm and observe the response to a requested increase in water level. Once observed return to 75mm.
5. Adjust the I term to 50 seconds in the PID controller. Observe the levels return faster than before.
6. Adjust setpoint to 100 mm and observe the response to a requested increase in water level. Once observed return to 75mm
7. Continue reducing the integral time setting and repeating the disturbances and setpoint until levels become unstable.
8. End of test

Experimental Challenges

- Reading the correct volume on the vessel when pumping in the water
- Cannot get accurate reading from the mimic diagram
- Knowing how much the pump speed should be set at to accurately maintain equilibrium
- Understanding the plotting of the graph when doing Exercise C

Safety

- Wear goggles
- Wear gloves (optional)
- Lab coat may be used to avoid any splash of water
- Must use DI water so the equipment does not malfunction
- Water can cause a hazard if it comes in contact with the outlet plug

Comment Summary

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